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FORM 2
THE PATENTS ACT, 1970
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AND
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COMPLETE SPECIFICATION
(See section 10 and rule 13)

10

Title of the Invention

**A PROSTHETIC LEG EQUIPPED WITH MACHINE LEARNING
BASED REAL TIME ADJUSTMENT**

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The following Specification particularly describes and ascertains the nature of this invention and the manner in which it is to be performed.

FIELD OF THE INVENTION:

[002] The present invention relates to a prosthetic leg. More specifically, the invention relates to a prosthetic leg equipped with a machine learning-based real-time adjustment system.

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BACKGROUND OF THE INVENTION:

[003] In recent years, there have been significant improvements in the field of prosthetics, especially when it comes to enhancing mobility, comfort, and the overall quality of life for people with lower limb amputations. Traditional prosthetic legs have mainly used mechanical and passive parts that don't always mimic the natural movement of a human leg. Because of this, users often face difficulties such as discomfort, instability, and challenges in adjusting to different walking conditions. For instance, when walking on uneven surfaces, climbing stairs, or changing walking speeds, traditional prosthetic limbs may not adapt smoothly or automatically to these varying situations. As a result, users may find themselves struggling with their prosthetic legs, feeling less confident or secure while walking. These limitations can significantly impact the user's independence, comfort, and overall mobility, as the prosthetic does not provide the same fluid, natural movement that a biological limb does.

[004] Current prosthetic legs still have significant problems when it comes to adapting to different walking conditions and ensuring user comfort. One of the biggest issues is that most prosthetics cannot automatically adjust when the user changes their walking environment, like walking on different types of terrain, changing speed, or going up or down stairs. Instead, these devices usually require manual adjustments or switching to preset

modes for different activities, which can make walking feel awkward, uncomfortable, and less natural. Additionally, many prosthetics don't learn from how the user walks, so they provide a one-size-fits-all response that doesn't match the user's specific movements. This leads to problems like instability, discomfort, and more fatigue while walking.

[005] Many existing prosthetics are also quite complicated and don't respond quickly enough to provide a smooth walking experience. Although some use sensors to detect movement, they often can't make immediate adjustments based on how the user walks or changes in the environment. This makes it harder for users to walk on different surfaces or adjust their speed easily. Additionally, these prosthetics tend to be heavy, use a lot of power, and aren't always durable, which can make them uncomfortable and difficult to use. The mechanical parts can wear out over time and need frequent repairs, making them expensive to maintain. These issues show the need for a more advanced prosthetic that can adapt in real-time to the user's movements and surroundings.

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OBJECT OF THE INVENTION:

[006] The principal object of the invention is to provide an auto-adjustable prosthetic leg that leverages machine learning and real-time sensory feedback to emulate the natural movements of the user's healthy leg.

[007] An objective of the present invention is to develop a prosthetic leg that improves user mobility and comfort by adapting to the user's gait and environmental conditions in real-time.

[008] Another object of the invention is to create a machine learning-based system that dynamically adjusts the prosthesis' behavior during different walking scenarios, such as varying speeds and terrain.

5 **[009]** Another object of the invention is to integrate sensors into the prosthetic leg to collect movement data, which will be used to predict and adjust the prosthesis' function in real-time.

[010] Another object of the invention is to develop a machine learning algorithm trained on a diverse dataset, enabling
10 the prosthesis to recognize and respond to different walking patterns and conditions.

[011] Another object of the invention is to enhance the stability, adaptability, and overall user satisfaction of prosthetic legs by offering a personalized, responsive, and intelligent
15 prosthetic experience.

[012] Another object of the invention is to provide individuals with lower limb amputations a more efficient and natural method of movement, improving their quality of life through advanced prosthetic technology.

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SUMMARY OF THE INVENTION:

[013] The present invention relates to a prosthetic leg with equipped with a machine learning based real time adjustment. The system comprises one or more sensors configured to capture
25 movement data of a user, a motor a memory includes a trained machine learning model. The system comprises at least one processor coupled with the one or more sensors, the memory, the motor.

[014] The at least one processor is configured to receive sensor data associated with detection a muscle flexion and muscle extension of a residual limb of a user. The at least one processor is further configured to train a ML model using a dataset of gait cycles of a plurality of users to obtain a trained ML model. The at least one processor is configured to predict an intended movement of the user using the sensor data and the trained ML model. The at least one processor is configured to control, via the motor, a movement of the prosthetic leg based on the predicted movement of the user.

[015] In an embodiment, the one or more sensors includes a flex sensor. The flex sensor is configured to detect muscle activity and movement characteristics of the user. The flex sensor is strategically placed on a residual limb of the user. The dataset of gait cycles includes walking, running, and climbing stairs. The at least one processor is further configured to analyze the sensor data in real-time and adjust the prosthetic leg's movement characteristics according to the user's gait, muscle strength, and activity level using the trained ML module.

[016] the processor is configured to adapt the prosthetic leg's performance to varying external conditions, including terrain type, slope, walking speed, and other environmental factors, based on continuous sensor input. the processor is configured to transmit, control signals to the motor that is embedded within the prosthetic leg and control, via the motor, adjustment of the prosthetic knee joint's angle of the prosthetic leg, mimicking natural leg movement and enabling a fluid gait.

[017] In an embodiment, the ML module is configured to determine adjustments to the prosthetic leg's joint angles,

actuation force, and damping characteristics to optimize comfort, stability, and mobility during use.

[018] The ML model stores historical sensor data that allows the ML module to personalize adjustments based on accumulated
5 usage patterns and user-specific physical characteristics over time.

BRIEF DESCRIPTION OF THE DRAWINGS:

[019] The present invention will hereinafter be described in
10 conjunction with the accompanying drawings, wherein like numerals denote like elements. Additional embodiments of the invention will become evident upon reviewing the non-limiting embodiments described in the specification in conjunction with the accompanying drawings, wherein:

15 **[020] Figures 1A and 1B** illustrate network environment for a prosthetic leg equipped with a machine learning-based real-time adjustment system, in accordance with an embodiment of the present invention.

20 **[021] Figures 2A, 2B, and 2C** illustrate a prosthetic leg equipped with a machine learning-based real-time adjustment system, in accordance with an embodiment of the present invention.

25 **[022] Figure 3** illustrates a system for monitoring and management of prosthetic leg, in accordance with an embodiment of the present invention.

[023] Figure 4 illustrates a method for monitoring and management of prosthetic leg, in accordance with an embodiment of the present invention.

[024] **Figure 5A** illustrates a flex sensor, in accordance with an embodiment of the present invention.

[025] **Figure 5B** illustrates a voltage division of the flex sensor, in accordance with an embodiment of the present
5 invention.

[026] **Figure 6** illustrates a servo motor, in accordance with an embodiment of the present invention.

[027] **Figure. 7** illustrates a method for monitoring and management of prosthetic leg, in accordance with an embodiment
10 of the present invention.

[028] **Fig.8A and 8B** illustrates a graph showing real positions of the user and the predicted positions using the system.

DETAILED DESCRIPTION OF THE PREFERRED
15 **EMBODIMENTS:**

[029] Before the present configuration of a prosthetic leg equipped with a machine learning-based real-time adjustment system, it is to be understood that this disclosure is not limited to particular assembly or configuration or arrangement for achieving
20 as described, since it may vary within the specification indicated. It is further to be understood that the terminology used in the description is only for the purpose of describing the particular versions or embodiments and is not intended to limit the scope of the present invention.

25 **[030]** The words, "comprising", "having", "including" & "containing" and other forms thereof, are intended to be equivalent in meaning and be open ended in that an item or items following any one of these words is not meant to be an exhaustive

listing of such item or items or meant to be limited to only the listed item or items.

[031] The present a prosthetic leg equipped with a machine learning-based real-time adjustment system, aiming to enhance mobility and comfort for users. The invention creates an intelligent prosthesis that adapts dynamically to the user's gait and environmental conditions. The methodology involves integrating sensors into the prosthetic leg to gather data on movement patterns, which are then processed by a machine learning algorithm to predict and adjust the prosthesis' behavior in real-time. The system is trained using a dataset collected from various users, enabling it to recognize and respond to different walking scenarios, such as changes in terrain or speed. The application of this technology extends to individuals with lower limb amputations, offering them a more personalized and responsive prosthetic experience. The machine learning-based real-time adjustment system represents a promising advancement in prosthetic technology, providing users with a more natural and efficient means of movement.

[032] **Figure 1A and 1B** illustrates network environment for a prosthetic leg equipped with a machine learning-based real-time adjustment system, in accordance with an embodiment of the present invention. Referring to drawings, the network environment (100) includes a prosthetic leg (101) (shown in Figure. 2A and 2B). The prosthetic leg (101) is equipped with a machine learning-based real-time adjustment system, aiming to enhance mobility and comfort for users. The prosthetic leg (101) is an artificial device designed to replace a leg that has been amputated or is non-functional due to injury, illness, or congenital

conditions of a user. The prosthetic leg (101) may include a socket that fits over the remaining portion of the leg (the residual limb). The socket is custom-molded to fit the individual's shape for comfort and stability. The prosthetic leg (101) may further include
5 pylon or shank that connects the socket to the foot, often made of lightweight materials like carbon fiber, aluminum, or titanium. The prosthetic leg (101) may further include a knee unit that allows for bending and straightening, enabling more natural movement. The unit can be mechanical or motorized for enhanced
10 functionality. The prosthetic leg (101) may include a foot. The foot makes contact with the ground. Prosthetic feet come in various designs, such as those that mimic the function of a human foot or advanced models with shock absorption and energy return mechanisms for more dynamic movement. In some example
15 embodiment, the prosthetic leg (101) may include suspension system that keeps leg securely in place on the residual limb, often using straps, suction, or a locking mechanism. The prosthetic leg (101) disclosed herein is an auto-adjustable prosthetic leg that leverages machine learning and real-time sensory feedback to
20 emulate the natural movements of the user's healthy leg.

[033] Referring back to Figure 1A and 1B, the prosthetic leg (101) includes a flux sensor (104) which is strategically placed to monitor the dynamics of the healthy leg of the user. The flux sensor (104) captures detailed movement patterns, which are
25 then processed by a machine learning algorithm to predict and replicate the movements in the prosthetic leg (101). The flex sensor (104) is a sensor that works on the principle of calculating the degree of deflection by the change of resistance. The sensor is inserted with the surface and the resistance built in the sensor

is varied by the flex, the flex signals in there are analog in nature. Which works as a voltage divider. Inside the flex sensor, there are carbon-resistive elements with a thin flexible substrate as more carbon leads to less resistance. As substrates bend the sensor (104) produces a resistance correlated to the output to the bend radius as shown in Figure 5B. The variation of the flex results leads to the variation of the resistance on its own. The main control unit further reads these resistance changes and it's given to ADC for converting these values into equivalent digital values.

10 **[034]** The prosthetic leg (101) further includes a servo motor (102), controlled by a processor such as an Arduino microcontroller, to adjust its position and movement in real time based on the input from the flux sensor (104). The servo motor (102) (illustrated in FIG. 6) is used as a rotatory controller that
15 gives precise control of angular posture for the prosthetic leg (101). Dissimilar to a DC motor, which just spins co continuously when power is being transferred, a servo motor can be informed about specific angles along with holding those positions. This is pulled off through a combination of a gear, a motor, a reduction
20 unit, a position sensor, and a control circuit. The control circuit received a PWM signal that notifies the specific position of the servo motor. The signals are further transmitted by a control circuit to compare the signals with feedback received from the position sensor to determine whether the rotation of the motor will
25 be clockwise or anticlockwise for the desired position. Furthermore, the motor is turned on up to the position sensor, which matches the feedback with the desired position signal.

[035] In an example embodiment, the prosthetic leg (101) may include a system (120). The system (120) may further

include a machine learning (ML) based real-time adjustment system. The system (120) further includes convolution neural networks (CNNs) and recurrent neural networks (RNNs). Machine learning algorithms, pre-loaded onto the Arduino, analyses the signal patterns. These algorithms are trained on a dataset of various gait cycles, including walking, running, and climbing stairs. By analyzing the flex sensor data and comparing it to the training data, the machine learning model can predict the user's intended movement with high accuracy.

10 **[036]** This integration of machine learning with adaptive hardware allows the prosthetic leg to continuously adjust to the user's gait, providing a more seamless and natural walking experience. By reducing the need for manual adjustments and improving the synchronization between the prosthetic and the user's natural movements, this technology has the potential to significantly enhance the autonomy and comfort of prosthetic leg users. In an example embodiment, the system may access the ML module (150) via a network (116). In one embodiment, the network (116) includes one or more networks such as a data network, a wireless network, a telephony network, or any combination thereof. It is contemplated that the data network may be any local area network (LAN), metropolitan area network (MAN), wide area network (WAN), a public data network (e.g., the Internet), short range wireless network, or any other suitable packet-switched network, such as a commercially owned, proprietary packet-switched network, e.g., a proprietary cable or fiber-optic network, and the like, or any combination thereof. In addition, the wireless network may be, for example, a cellular network and may employ various technologies including enhanced

data rates for global evolution (EDGE), general packet radio service (GPRS), global system for mobile communications (GSM), Internet protocol multimedia subsystem (IMS), universal mobile telecommunications system (UNITS), etc., as well as any other
5 suitable wireless medium, e.g., worldwide interoperability for microwave access (WiMAX), Long Term Evolution (LTE) networks, code division multiple access (CDMA), wideband code division multiple access (WCDMA), wireless fidelity (Wi-Fi), wireless LAN (WLAN), Bluetooth®, Internet Protocol (IP) data casting, satellite,
10 mobile ad-hoc network (MANET), and the like, or any combination thereof.

[037] In an embodiment, the network environment (120) may include a cloud-based storage (110). In an embodiment, the all the collected is stored in the cloud-based storage (110). In an
15 embodiment, the cloud-based storage (110) stores historical data for analysis and to operation of the prosthetic leg (101).

[038] **Figure 3** illustrates a system for monitoring and management of prosthetic leg, in accordance with an embodiment of the present invention. ESP 8266: The system may be an
20 ESP8266 that is a popular low-cost Wi-Fi microchip developed by express if Systems, widely used in Internet of Things (IoT) applications due to its affordability and versatility. It features a built-in 32-bit low-power microcontroller, integrated Wi-Fi supporting IEEE 802.11 b/g/n standards, and multiple GPIO pins
25 for interfacing with various sensors and actuators. The chip can be programmed using environments like the Arduino IDE and Lua, making it accessible for hobbyists and developers. Available in various module forms, such as the ESP-01 and NodeMCU, the ESP8266 is a staple in home automation, sensor networks, and

DIY electronics projects. Alternatively, the system includes a processor (210), a memory (220), and a network interface (230). In an embodiment, the processor (210) may be Arduino is one of the microcontrollers. The processor (210) may also be
5 accompanied with one or more specialized components to perform certain processing functions and tasks such as one or more digital signal processors (DSP), or one or more application-specific integrated circuits (ASIC). A DSP typically is configured to process real-world signals (e.g., sound) in real time independently of the
10 processor. Similarly, an ASIC can be configured to performed specialized functions not easily performed by a general purposed processor. Other specialized components to aid in performing the inventive functions described herein include one or more field programmable gate arrays (FPGA) (not shown), one or more
15 controllers (not shown), or one or more other special-purpose computer chips. The processor (210) and accompanying components have connectivity to the memory (220) via the bus. In an embodiment, the processor (210) may be an ESP32 microcontroller. The ESP32 microcontroller is a powerful, low-cost
20 microcontroller with integrated Wi-Fi and Bluetooth capabilities, making it ideal for Internet of Things (IoT) applications. The ESP32 can be programmed using various platforms, including the Arduino IDE, Espressif's ESP-IDF (IoT Development Framework), and MicroPython. Data IOT projects logging, environmental
25 monitoring, and remote-control applications.

[039] The memory (220) includes both dynamic memory (e.g., RAM, magnetic disk, writable optical disk, etc.) and static memory (e.g., ROM, CD-ROM, etc.) for storing executable instructions that when executed perform the inventive steps

described herein to recommending video content. The memory (220) also stores the data associated with or generated by the execution of the inventive steps.

[040] Network interface (230) provides a one-way or two-way communication coupling to a variety of external devices that operate with their own processors, such as printers, scanners and external disks. In general, the coupling is with a network link that is connected to a local network to which a variety of external devices with their own processors are connected. For example, network interface may be a parallel port or a serial port or a universal serial bus (USB) port on a personal computer. In some embodiments, network interface is an integrated services digital network (ISDN) card or a digital subscriber line (DSL) card or a telephone modem that provides an information communication connection to a corresponding type of telephone line. In some embodiments, a network interface is a cable modem that converts signals on bus into signals for a communication connection over a coaxial cable or into optical signals for a communication connection over a fiber optic cable.

[041] **Figure 4** illustrates a method for controlling prosthetic leg. The method includes at step 401, receiving sensor data associated with detection a muscle flexion and muscle extension of a residual limb of a user. The sensors data is received from a flex sensor (104). The flex sensor (104) is configured to detect muscle activity and movement characteristics of the user. The flex sensor (104) is strategically placed on a residual limb of the user. The dataset of gait cycles includes walking, running, and climbing stairs.

[042] In an embodiment, the method further includes at step 403, training a ML model (150) using a dataset of gait cycles of a plurality of users to obtain a trained ML model. For real-time gait analysis and potential remote rehabilitation guidance, the processor (210) transmits the processed sensor data (including raw sensor values, predicted movement, and motor control signals) to a server via a Wi-Fi module attached to the board. The server can be a cloud-based platform or a local computer. On the server-side, another machine learning model can further analyze the received data. This model identifies gait abnormalities, potential falls, or areas for improvement in the user's walking pattern. The analysis results can be visualized on a user interface accessible through a smartphone or computer, providing valuable feedback for the user and their therapist. Additionally, the therapist can remotely monitor the user's progress, adjust the machine learning algorithms on the server, and personalize the prosthetic leg's behavior for optimal performance.

[043] In an embodiment, the method includes at step 405, predicting an intended movement of the user using the sensor data and the trained ML model (150). By analyzing the flex sensor data and comparing it to the training data, the machine learning model (150) may predict the user's intended movement with high accuracy. Based on the predicted movement and the user's pre-defined settings, the processor (210) sends control signals to a servo motor (102) embedded within the prosthetic leg (101). The system (120) herein uses a Linear Regression model for Predicting the banding angle of the prosthetic leg (101). Along with Linear Regression the system also uses Supervision based learning for better prediction. If the banning angle of Healthy leg is less than

45° it will follow linear regression model if it is greater than 45° it will follow Supervision based learning.

[044] In an embodiment, the method further includes at step 407, controlling, via the motor, a movement of the prosthetic leg (101) based on the predicted movement of the user. The motors (102) adjust the prosthetic knee joint's angle, mimicking natural leg movement and enabling a more fluid gait. the processor is configured to adapt the prosthetic leg's performance to varying external conditions, including terrain type, slope, walking speed, and other environmental factors, based on continuous sensor input. The ML module (105) stores historical sensor data that allows the ML module to personalize adjustments based on accumulated usage patterns and user-specific physical characteristics over time. 9. The prosthetic leg as claimed in claim 1, wherein the system includes a safety mechanism that adjusts or disables the prosthetic leg's response when abnormal sensor data is detected, such as a fall or abnormal gait pattern, to prevent injury. The system (120) is configured to integrate with external devices such as smartphones, rehabilitation tools, or smart home systems to provide remote monitoring, control, and feedback to enhance the user's experience and rehabilitation process.

[045] The system (120) disclosed herein offers several advantages. The flex sensors (104) provide a comfortable and natural way to control the prosthetic leg, mimicking how the brain communicates with biological muscles. Machine learning on the Arduino allows for real-time movement prediction and efficient control, while server-side analysis with another machine learning model enables detailed gait assessment and potential remote rehabilitation guidance. The Wi-Fi module facilitates data

transmission, creating a connected prosthetic system that can evolve and improve over time.

[046] Results for accuracy prediction:

[047] Fig.8A and 8B illustrates a graph showing real
5 positions of the user and the predicted positions using the system
(120). A total 200 data set shown in Fig 8A and 8B. Fig. 8A
illustrates the real position of the user. The Fig. 8b illustrates the
predicted results using the system (120). The real positions and
predicted positions are matching from the Fig 8A and 8B.

10 **[048]** Technical advantages of the invention:

- Real-Time Movement Prediction: The integration of machine learning on the Arduino allows for immediate prediction and control of prosthetic movements, enabling smooth and responsive operation, similar to natural limb motion.
- 15 • Efficient Control with Flex Sensors: Flex sensors provide a comfortable and intuitive control method for the user, closely mimicking how biological muscles interact with the brain, offering more natural and adaptive movement.
- Remote Rehabilitation and Adjustment: The cloud
20 computing and machine learning system enable remote gait assessment and adjustments, allowing for continuous monitoring and rehabilitation guidance, which can improve the user's experience and health outcomes.
- Adaptability and Continuous Improvement: The system's
25 Wi-Fi connectivity allows for continuous data transmission, enabling updates and improvements to the prosthetic system over time, which helps the device evolve based on the user's needs.

- Machine Learning-Enhanced Gait Assessment: Server-side machine learning models provide detailed analysis of the user's gait, helping clinicians or rehabilitation professionals offer personalized adjustments and interventions for better prosthetic use.
- Scalability and Versatility: The combination of sensor technologies and cloud-based machine learning allows for easy adaptation to different users, making the prosthetic leg suitable for a wide range of individuals and diverse mobility needs.
- Improved Quality of Life: The use of machine learning for real-time adjustments and the potential for remote monitoring ensures that the prosthetic can evolve in response to the user's behavior and environment, significantly enhancing overall user comfort and mobility.

[049] Although the subject matter has been described in language specific to structural features and/or methods in considerable detail with reference to certain preferred embodiments thereof, it is to be understood that the implementations and/or embodiments are not necessarily limited to the specific features or methods described. The examples described in detail here are only some possible embodiments of the invention among others and it could be subjected to many alterations and variants within the grasp of those skilled in the art. As such, the spirit and scope of the appended claims should not be limited to the description of the preferred embodiments contained therein.

WE CLAIM:

1. A prosthetic leg with equipped with a machine learning based real time adjustment, comprising:

5 one or more sensors (108) configured to capture movement data of a user;

 a motor (102);

 a memory (220) includes a trained machine learning model (150);

10 at least one processor (210) coupled with the one or more sensors (108), the memory (220), the motor (102), wherein the at least one processor (210) is configured to:

 receive sensor data associated with detection a muscle flexion and muscle extension of a residual limb of a user;

15 train a ML model (150) using a dataset of gait cycles of a plurality of users to obtain a trained ML model (150);

 predict an intended movement of the user using the sensor data and the trained ML model (150); and

20 control, via the motor (102), a movement of the prosthetic leg (101) based on the predicted movement of the user.

2. The prosthetic leg (101) as claimed in claim 1, wherein the one or more sensors includes a flex sensor (104).

25 **3.** The prosthetic leg (101) as claimed in claim 2, wherein the flex sensor (104) is configured to detect muscle activity and movement characteristics of the user.

4. The prosthetic leg (101) as claimed in claim 2, wherein the flex sensor (104) is strategically placed on a residual limb of the user.

5. The prosthetic leg (101) as claimed in claim 1, wherein the
5 dataset of gait cycles includes walking, running, and climbing stairs.

6. The prosthetic leg (101) as claimed in claim 1, wherein the at
least one processor (210) configured to analyze the sensor data
10 in real-time and adjust the prosthetic leg's movement characteristics according to the user's gait, muscle strength, and activity level using the trained ML module (105).

7. The prosthetic leg as claimed in claim 1, wherein the processor
15 is configured to adapt the prosthetic leg's performance to varying external conditions, including terrain type, slope, walking speed, and other environmental factors, based on continuous sensor input.

8. The prosthetic leg as claimed in claim 1, wherein the processor
20 is configured to:

transmit, control signals to the motor that is embedded within
the prosthetic leg (101); and

control, via the motor, adjustment of the prosthetic knee
25 joint's angle of the prosthetic leg (101), mimicking natural leg movement and enabling a fluid gait.

9. The prosthetic leg as claimed in claim 1, wherein the ML module is configured to determine adjustments to the prosthetic leg's joint angles, actuation force, and damping characteristics to optimize comfort, stability, and mobility during use.

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10. The prosthetic leg as claimed in claim 1, wherein the ML model (150) stores historical sensor data that allows the ML module (150) to personalize adjustments based on accumulated usage patterns and user-specific physical characteristics over time.

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ABSTARCT
**A PROSTHETIC LEG EQUIPPED WITH MACHINE LEARNING
BASED REAL TIME ADJUSTMENT**

5 The present invention relates to a prosthetic leg equipped with a machine learning-based real-time adjustment system. The system (120) includes one or more sensors (108), a motor (102), a memory (220), and at least one processor (210). The at least one processor (210) is configured to receive sensor data associated
10 with detection a muscle flexion and muscle extension of a residual limb of a user. The at least one processor (210) is further configured to train a ML model (150) using a dataset of gait cycles of a plurality of users. The at least one processor (210) is further configured to predict an intended movement of the user using the
15 sensor data and the trained ML model (150) and control, via the motor (102), a movement of the prosthetic leg (101) based on the predicted movement of the user. **Figure 1 A.**